

1. CHECK S3 BUCKETS (STORAGE LAYER)

Check Bucket Security

```
akosa@AKOSA:~$ aws s3api get-public-access-block --bucket samuel-employee-images
{
  "PublicAccessBlockConfiguration": {
    "BlockPublicAcls": true,
    "IgnorePublicAcls": true,
    "BlockPublicPolicy": true,
    "RestrictPublicBuckets": true
  }
}
akosa@AKOSA:~$ |
```

Check Versioning

```
akosa@AKOSA:~$ aws s3api get-bucket-versioning --bucket samuel-employee-images
{
  "Status": "Enabled"
}
akosa@AKOSA:~$ |
```

2. CHECK LAMBDA FUNCTIONS (COMPUTE LAYER)

List Functions

```
akosa@AKOSA:~$ aws lambda list-functions --query "Functions[].FunctionName"
[
  "employee-authentication",
  "employee-registration"
]
akosa@AKOSA:~$ |
```

Check Configuration

```
akosa@AKOSA:~$ aws lambda get-function-configuration \
--function-name employee-authentication
{
  "FunctionName": "employee-authentication",
  "FunctionArn": "arn:aws:lambda:eu-west-1:009850210027:function:employee-authentication",
  "Runtime": "python3.14",
  "Role": "arn:aws:iam::009850210027:role/FacialRecognitionLambdaRole-sam",
  "Handler": "lambda_function.lambda_handler",
  "CodeSize": 1054,
  "Description": "",
  "Timeout": 60,
  "MemorySize": 500,
  "LastModified": "2026-04-10T16:02:21.000+0000",
  "CodeSha256": "sgfDrD5opj1QdtubrM+hBz+8xtG4dq9KYotdGqaQ5XU=",
  "Version": "$LATEST",
  "TracingConfig": {
    "Mode": "PassThrough"
  },
  "RevisionId": "1758f501-c5f2-4e72-8987-db80717ce226",
  "State": "Active",
  "LastUpdateStatus": "Successful",
  "PackageType": "Zip",
  "Architectures": [
    "x86_64"
  ],
  "EphemeralStorage": {
    "Size": 512
  },
  "SnapStart": {
    "ApplyOn": "None",
    "OptimizationStatus": "Off"
  },
  "RuntimeVersionConfig": {

```

Test Lambda manually

```
akosa@AKOSA:~$ aws lambda invoke \
--function-name employee-authentication \
--payload '{}' \
response.json && cat response.json
{
  "StatusCode": 200,
  "ExecutedVersion": "$LATEST"
}
{"statusCode": 400, "headers": {"Access-Control-Allow-Origin": "*", "Access-Control-Allow-Headers": "*"}, "body": "\"Missing image\""}
akosa@AKOSA:~$
```

3. CHECK API GATEWAY (API LAYER)

List APIs

```
akosa@AKOSA:~$ aws apigateway get-rest-apis \
--query "items[].name"
[
  "facial-recognition-api"
]
akosa@AKOSA:~$
```

Get API Gateway ID for our API

```
akosa@AKOSA:~$ aws apigateway get-rest-apis \
--query "items[?name=='facial-recognition-api'].id" \
--output text
h3v6z64lxd
akosa@AKOSA:~$ |
```

Check Resources

```
akosa@AKOSA:~$ export API_ID=h3v6z64lxd
akosa@AKOSA:~$ aws apigateway get-resources \
--rest-api-id $API_ID \
--region eu-west-1
{
  "items": [
    {
      "id": "iq3hgn",
      "parentId": "qyttegxb6j",
      "pathPart": "bucket",
      "path": "/bucket",
      "resourceMethods": {
        "POST": {}
      }
    },
    {
      "id": "qm3bu8",
      "parentId": "qyttegxb6j",
      "pathPart": "upload",
      "path": "/upload",
      "resourceMethods": {
        "OPTIONS": {},
        "POST": {}
      }
    },
    {
      "id": "qyttegxb6j",
      "path": "/"
    }
  ]
}
akosa@AKOSA:~$ |
```

Resource ID

```
akosa@AKOSA:~$ aws apigateway get-resources \
--rest-api-id $API_ID \
--query "items[?path=='/bucket'].id" \
--output text
iq3hgn
akosa@AKOSA:~$
```

Check Method (POST)

```
akosa@AKOSA:~$ aws apigateway get-method \
--rest-api-id h3v6z64lxd \
--resource-id iq3hgn \
--http-method POST \
--region eu-west-1
{
  "httpMethod": "POST",
  "authorizationType": "NONE",
  "apiKeyRequired": true,
  "requestParameters": {},
  "requestModels": {},
  "methodResponses": {
    "200": {
      "statusCode": "200",
      "responseModels": {
        "application/json": "Empty"
      }
    }
  },
  "methodIntegration": {
    "type": "AWS",
    "httpMethod": "POST",
    "uri": "arn:aws:apigateway:eu-west-1:lambda:path/2015-03-31/functions/arn:aws:lambda:eu-west-1:009850210027:function:employee-authentication/invocations",
    "passthroughBehavior": "WHEN_NO_MATCH",
    "contentHandling": "CONVERT_TO_TEXT",
    "timeoutInMillis": 29000,
    "cacheNamespace": "iq3hgn",
    "cacheKeyParameters": [],
    "integrationResponses": {
      "200": {
        "statusCode": "200",
        "responseTemplates": {

```

Lambda Integration

```
akosa@AKOSA:~$ aws apigateway get-integration \
--rest-api-id h3v6z64lxd \
--resource-id iq3hgn \
--http-method POST \
--region eu-west-1
{
  "type": "AWS",
  "httpMethod": "POST",
  "uri": "arn:aws:apigateway:eu-west-1:lambda:path/2015-03-31/functions/arn:aws:lambda:eu-west-1:009850210027:function:employee-authentication/invocations",
  "passthroughBehavior": "WHEN_NO_MATCH",
  "contentHandling": "CONVERT_TO_TEXT",
  "timeoutInMillis": 29000,
  "cacheNamespace": "iq3hgn",
  "cacheKeyParameters": [],
  "integrationResponses": {
    "200": {
      "statusCode": "200",
      "responseTemplates": {
        "application/json": null
      }
    }
  },
  "responseTransferMode": "BUFFERED"
}
```

Check Deployment Stages

```
akosa@AKOSA:~$ aws apigateway get-stages \
--rest-api-id h3v6z64lxd
{
  "item": [
    {
      "deploymentId": "pdqmof",
      "stageName": "dev",
      "cacheClusterEnabled": false,
      "cacheClusterStatus": "NOT_AVAILABLE",
      "methodSettings": {},
      "tracingEnabled": false,
      "createdDate": "2026-04-10T15:44:43+03:00",
      "lastUpdatedDate": "2026-04-10T19:04:02+03:00"
    },
    {
      "deploymentId": "2yd0z9",
      "stageName": "v1",
      "cacheClusterEnabled": false,
      "cacheClusterSize": "0.5",
      "cacheClusterStatus": "NOT_AVAILABLE",
      "methodSettings": {},
      "tracingEnabled": false,
      "createdDate": "2026-04-10T15:16:00+03:00",
      "lastUpdatedDate": "2026-04-14T11:46:11+03:00"
    }
  ]
}
```

akosa@AKOSA:~\$

Get API Key ID and Value (used for authentication)

```
akosa@AKOSA:~$ aws apigateway get-api-keys \
--include-values \
--query "items[].{ID:id,Name:name,Key:value}"
[
  {
    "ID": "i6j65e0nv8",
    "Name": "facial-recognition-app-sam",
    "Key": "X6XjW8u39h5b0TC5EKJHu1AbQJNgNdyQ4hS6UmH8"
  }
]
```

akosa@AKOSA:~\$

Usage Plan ID

```
akosa@AKOSA:~$ aws apigateway get-usage-plans \
--query "items[].{id:id,name:name}"
[
  {
    "id": "4z7vj7",
    "name": "facial-recognition-sam"
  }
]
akosa@AKOSA:~$ |
```

Test Endpoint

```
akosa@AKOSA:~$ curl -X POST \
https://h3v6z64lxd.execute-api.eu-west-1.amazonaws.com/v1/bucket \
-H "Content-Type: application/json" \
-H "x-api-key: X6XjW8u39h5b0TC5EKJHu1AbQJNgNdyQ4hS6UmH8" \
-d '{"firstName":"Test","lastName":"User","image":"base64string"}'
{"statusCode": 400, "headers": {"Access-Control-Allow-Origin": "*", "Access-Control-Allow-Headers": "*"}, "body": "\"Missing image\""}
akosa@AKOSA:~$ |
```

4. CHECK RECOGNITION (AI LAYER)

List Collections

```
akosa@AKOSA:~$ aws rekognition list-collections
{
  "CollectionIds": [
    "employees"
  ],
  "FaceModelVersions": [
    "7.0"
  ]
}
akosa@AKOSA:~$ |
```

Describe Collection

```
akosa@AKOSA:~$ aws rekognition describe-collection \
--collection-id employees
{
  "FaceCount": 12,
  "FaceModelVersion": "7.0",
  "CollectionARN": "arn:aws:rekognition:eu-west-1:009850210027:collection/employees",
  "CreationTimestamp": "2026-04-10T10:52:28.561000+03:00",
  "UserCount": 0
}
akosa@AKOSA:~$ |
```

5. CHECK DYNAMODB (DATABASE LAYER)

List Tables

```
akosa@AKOSA:~$ aws dynamodb list-tables
{
  "TableNames": [
    "employee"
  ]
}
akosa@AKOSA:~$ |
```

Scan Table

```
akosa@AKOSA:~$ aws dynamodb scan \
--table-name employee \
--limit 5
{
  "Items": [
    {
      "rekognitionid": {
        "S": "4ded56f9-52be-465f-b475-988c77d52c97"
      },
      "FirstName": {
        "S": "Auto"
      },
      "LastName": {
        "S": "Registered"
      },
      "ImageKey": {
        "S": "Jeff_Bezos.jpeg.jpg"
      }
    },
    {
      "rekognitionid": {
        "S": "7b934aa1-73c2-4bef-b4be-62d985fcfd5a"
      },
      "FirstName": {
        "S": "Auto"
      },
      "LastName": {
        "S": "Registered"
      },
      "ImageKey": {
        "S": "Sergey-Brin.jpeg.jpg"
      }
    }
  ],
}
```


6. CHECK IAM ROLES (SECURITY LAYER)

Check Lambda Role

```
akosa@AKOSA:~$ aws lambda get-function-configuration \
--function-name employee-authentication \
--region eu-west-1 \
--query "Role"
"arn:aws:iam::009850210027:role/FacialRecognitionLamdaRole-sam"
akosa@AKOSA:~$
```

7. CHECK CLOUDWATCH (OBSERVABILITY)

List Log Groups

```
akosa@AKOSA:~$ aws logs describe-log-groups \
--query "logGroups[].logGroupName"
[
  "/aws/lambda/employee-authentication",
  "/aws/lambda/employee-registration",
  "RDSOSMetrics"
]
akosa@AKOSA:~$
```

Tail Logs

```
akosa@AKOSA:~$ aws logs tail /aws/lambda/employee-authentication \
--region eu-west-1 \
--follow
2026-04-14T08:46:28.250000+00:00 2026/04/14/[$LATEST]49a7d7a8dba7492baf90c705de78ee79 INIT_START Runtime Version: python:3.14.v38 R
untime Version ARN: arn:aws:lambda:eu-west-1::runtime:6198fb9c708e48010232459840699b2264ba8cc36621c703d5cd31b2245e02d1
2026-04-14T08:46:29.028000+00:00 2026/04/14/[$LATEST]49a7d7a8dba7492baf90c705de78ee79 START RequestId: 7fcb23e3-3644-45ba-908b-45f63b
7359b9 Version: $LATEST
2026-04-14T08:46:29.030000+00:00 2026/04/14/[$LATEST]49a7d7a8dba7492baf90c705de78ee79 END RequestId: 7fcb23e3-3644-45ba-908b-45f63b73
59b9
2026-04-14T08:46:29.031000+00:00 2026/04/14/[$LATEST]49a7d7a8dba7492baf90c705de78ee79 REPORT RequestId: 7fcb23e3-3644-45ba-908b-45f63
b7359b9 Duration: 2.14 ms Billed Duration: 776 ms Memory Size: 500 MB Max Memory Used: 100 MB Init Duration: 773.62 ms
```